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Treatment of therapy-resistant OCD with application of transcranial alternating current stimulation (tACS): a case series

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Abstract

Background Obsessive-compulsive disorder (OCD) is a chronic condition that significantly impacts patients' lives. Various treatments, including pharmacological, behavioral, and brain stimulation methods, are employed. Recent studies indicate that transcranial alternating current stimulation (tACS) effectively alleviates OCD symptoms, warranting further investigation into its therapeutic potential.

Methods A 4-week study at the Tertiary Center of Psychiatry and Behavioral Science in Southern Iran evaluated the effectiveness of tACS for treating OCD in participants aged 18–60. The treatment involved bi-weekly sessions targeting frontal brain areas, assessing various psychological metrics. Results indicated a significant improvement in specific measures. All participants met predefined therapy-resistance criteria based on documented non-response to evidence-based pharmacotherapy and/or structured CBT prior to enrolment.

Results Ninety-five patients were screened, and five eligible participants with OCD (ages 18–46; four female) completed the intervention. All had severe baseline symptoms (Y-BOCS ≥ 23). Obsessive-compulsive symptoms showed consistent improvement, with mean Y-BOCS decreasing from 28.4 ± 4.16 to 18.4 ± 6.34 (35.2% reduction). CGI-S scores declined from 5.4 ± 0.89 to 4.6 ± 1.14 , and the mean CGI-I score was 2.6, indicating notable clinical improvement. Affective symptoms improved substantially: total DASS-21 scores decreased by 46.4%, including reductions in stress, depression, and anxiety. SBQ-R scores also declined (–16.7%). Cognitive performance showed generally favorable trends, with improvement in TOLT planning scores. No major adverse effects were reported, aside from mild irritation and brief flashing sensations.

Conclusions Our pilot study on therapy-resistant OCD with tACS yields promising initial results and safety. However, limitations stress cautious interpretation, necessitating larger, controlled trials for validation and refinement.

Clinical trial number Not applicable.

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Keywords OCD, tACS, Therapy-Resistant OCD, Noninvasive brain stimulation, Psychiatry

Introduction

Obsessive-compulsive disorder (OCD) is a chronic psychiatric disorder with a prevalence of 1% of the world population [1]. OCD is characterized by obsessions and compulsions which are repetitive thoughts and actions respectively [2]. It contains a variety of symptoms including anxiety, and depression symptoms which are common in these patients [3, 4]. To alleviate these symptoms pharmacological and cognitive-behavioral therapies are effective to some extent and improve the quality of life in these patients [5].

OCD treatment mainly consists of pharmacological or cognitive-behavioral therapies or a combination of these approaches. Antidepressant drugs including selective serotonin reuptake inhibitors are the most common pharmacological treatment used in OCD. Other pharmacologic treatments such as antipsychotic drugs can also be used as add-on therapy in resistant types of OCD [6]. In addition, due to different responses to the pharmacological treatment of OCD, a personalized approach is recommended [7]. However, despite all improvements in the treatment of OCD in the last decades, still 40–60% of patients suffer from residual symptoms and the resistant types of the disease do not fully respond to the common treatments. Therefore, new strategies for the treatment of OCD are required [8].

Non-invasive transcranial brain stimulation (NTBS) therapies have been proposed as new strategies in the treatment of psychiatric disorders. tACS is a NTBS which delivers alternating electrical currents to the scalp. It involves a sinusoidal waveform current that reverses the direction of electron flow rhythmically every half-cycle and affects brain oscillations [9]. The neural activity of the brain consists of rhythmic patterns, which are called brain oscillations. These oscillations are required for normal function and connectivity of various brain regions [10].

It has been revealed that in psychiatric disorders including OCD these normal functioning brain oscillations are impaired. Specifically in the OCD it has been shown that impairments of the fronto-striatal loops occur in these patients [11]. Given the potential effects of tACS on cortical neurons and its synchronizing or desynchronizing effects on brain oscillations it has been considered as a promising effective treatment in the OCD [12].

Recently, clinical studies have investigated the effects of tACS in the psychiatry field and it has been shown to have promising effects in the treatment of psychiatric disorders including OCD [13, 14]. To the best of our knowledge, the studies evaluating the effects of tACS on OCD are case-reports with small number of participants

and the efficacy of this method on the treatment of different symptoms of OCD has not yet been determined with strong evidence. Therefore, in the current study, we investigated the effects of tACS on the symptoms of OCD in a case series study with the hope to pave the way to the treatment of this disease.

Methods

Study design

A four-week study was carried out at the Tertiary Center of Psychiatry and Behavioural Science in southern Iran between March and July 2023, covering the entire process from patient outreach to the completion of the intervention. The study was approved by the Shiraz University of Medical Sciences (SUMS) under the code [IR.SUMS.REC.1402.094]. For participant recruitment, individuals diagnosed with OCD who had been discharged within the past year from affiliated hospitals (Hafez and Ibn Sina clinics in Shiraz) were contacted. During the outreach, we called the potential participants and explained the study, including its purpose, potential benefits, possible risks, and available alternative treatments. Those meeting the inclusion criteria and willing to participate received tACS over a four-week period. The stimulation was integrated into their ongoing treatment plans without interrupting their usual care.

Participants

We enrolled participants 18–60 years old who had suffering from OCD, referring to Hafez and Ibn Sina Hospital clinics in Shiraz. The stimulation test was added to the treatment, and the patients' current treatment was not disrupted.

Inclusion criteria

As per the guidelines delineated in the Diagnostic and Statistical Manual-V (DSM-V), eligibility criteria for prospective study participants encompass a confirmed diagnosis of OCD substantiated by a qualified psychiatrist. In addition, prospective participants should fall within the age bracket of 18 to 60 years, demonstrate a scoring threshold of 23 or above on the Yale-Brown Obsessive Compulsive Scale (Y-BOCS) (indicative of individuals grappling with severe OCD symptoms), exhibit the absence of concurrent psychiatric and neurological disorders, possess the requisite cognitive and physical aptitude to engage in behavioral assessments and questionnaire completion, and manifest a voluntary commitment to partake in the research endeavor.

Exclusion criteria

Inconsistencies pertaining to the adherence with the established study inclusion criteria, reluctance on the part of patients to engage in study participation, incapacity of patients to undergo diagnostic examinations, the presence of comorbidities related to Axis I psychiatric disorders, ineligibility for tACS owing to the presence of metal implants, a historical record of seizures, head trauma, or neurosurgical procedures, prior receipt of electroconvulsive therapy (ECT), repetitive transcranial magnetic stimulation (rTMS), or transcranial direct current stimulation (tDCS), participants afflicted with myasthenia gravis, those with implanted devices such as pacemakers, mechanical heart valves, aneurysm clips, hip replacements, or any other metallic implant inadvertently introduced into their anatomy, individuals currently experiencing a significant medical ailment, a heightened risk of suicidal tendencies, and female subjects who are either pregnant, of childbearing age, sexually active without employing reliable contraception methods, or presently engaged in breastfeeding, shall be ineligible for inclusion in the study.

Intervention

The scalp was fitted with electrodes using Ten20 paste (Bio-Medical Instruments, Clinton Township, Michigan). Stimulation was delivered using a Starstim® multi-channel stimulator (Neuroelectronics, Barcelona, Spain), a CE-certified, battery-powered system supporting programmable multi-site tACS protocols. Two Ag/AgCl PiStim electrodes (1 cm radius, contact area $\approx 3.14 \text{ cm}^2$) were positioned over the left and right frontal areas (F3 and F4, respectively, in the 10–20 system), with a third $5 \times 7 \text{ cm}$ “return/reference” electrode placed over the vertex (Cz) using a PiStim-type sponge electrode. This setup ensures synchronized stimulation of both frontal cortices, targeting frontal alpha/gamma imbalance implicated in OCD. The electrode montage ($5 \times 5 \text{ cm}$ over F3/F4, $5 \times 7 \text{ cm}$ return at Cz) draws from prior tACS studies targeting bilateral frontal regions to entrain gamma rhythms. This three-electrode dual-site configuration allows controlled current flow between the two active frontal sites and a neutral return, enhancing focality and network-level coupling. Using a larger reference electrode at Cz helps optimize current flow through frontal cortices while maintaining comfort and safety [15].

Participants received biweekly tACS (twice per week for four weeks, eight sessions total) to maintain therapeutic dosing while prioritizing tolerability. User experiences and community guidance suggest that one to two sessions per week are generally safe and better tolerated than more frequent schedules [16, 17]. The stimulation protocol used a 40 Hz (gamma-band) sine wave, with amplitude set to 2 mA at Cz and 1 mA at F3 and F4

(zero-to-peak). The 40 Hz frequency is based on evidence demonstrating its ability to entrain frontal cortical oscillations, improve cognitive control, and reduce psychiatric symptoms, including in OCD (18). Notably, a multi-session studies demonstrated improvements in working memory and processing speed following of 40 Hz tACS over F3/F4 in both depressed and OCD populations, and user-reported anecdotal benefits from similar montages/channels reinforce its feasibility [12, 18]. Targeting F3/F4 in tACS (or tDCS) is therefore strongly justified both theoretically (CSTC loop modulation) and empirically (clinical improvement and electrophysiological normalization) [19, 20].

Each session included a 20-second ramp-in, 20 min of continuous 40 Hz tACS, and a 20-second ramp-out (total 1,160 s). Participants sat upright with eyes open, viewing a ReefScapes video to mask phosphene effects and maintain a consistent attentional state.

Side effects and safety

Following every session, we promptly documented the individual's encountered side effects and any complications that arose within the span of these four hours through self-reports.

Outcome

The primary outcome was the Y-BOCS, with key secondary outcomes including the Clinical Global Impression–Severity (CGI-S) and Clinical Global Impression–Improvement (CGI-I). Exploratory outcomes, analyzed descriptively, comprised the DASS-21 (total and subscales), Suicide Behaviors Questionnaire-Revised (SBQ-R), Iowa Gambling Task (IGT), and Tower of London Test (TOLT). All assessments were obtained at baseline and weekly through week 4, and exploratory measures are summarized descriptively in the Supplementary materials.

Demographic and clinical characteristic

Before initiating the treatment, patients were inquired about their demographic and clinical details, such as age, gender, age of disease onset, prescribed medications, and related information.

Yale Brown Obsessive-Compulsive Scale (Y-BOCS)

In our study, we employed the Persian iteration of the Y-BOCS assessment tool [21] to evaluate patients at the study's outset, as well as at the first, second, third, and fourth weeks. Y-BOCS is a semi-structured interview consisting of ten items, conducted by a trained clinician to ascertain the nature and intensity of OCD symptoms experienced by the patients within the previous seven days, with scores ranging from 0 to 4 [22].

Depression Anxiety and Stress Scale 21 (DASS-21)

The DASS-21 is a set of three subcategories for self-assessment that aim to evaluate feelings of depression, anxiety, and stress. Each of these subcategories contains seven questions. The questionnaire follows a four-choice Likert scale, ranging from “not at all” to “very much.” The scoring system assigns zero as the lowest score and three as the highest score, with “not at all” and “very much” corresponding to the respective extremes. In Iran, the questionnaire’s validity and reliability were verified, showing a Cronbach’s alpha coefficient of 0.77 for depression, 0.79 for anxiety, and 0.78 for stress [23].

The Suicide Behaviors Questionnaire-Revised (SBQ-R)

The assessment of suicide risk was carried out using the SBQ-R, which consisted of four items. These items inquired about lifetime experiences of suicidal thoughts, the frequency of such thoughts within the past year, any indications of a threat of a suicide attempt, and the self-reported likelihood of actually attempting suicide. A total score equal to or exceeding 7 on this questionnaire served as the cutoff point to identify individuals at risk of suicide [24].

The Clinical Global Impressions scale (CGI)

The CGI is a widely employed tool for evaluating the overall severity of and improvement in OCD symptoms [25]. In this context, an impartial evaluator assigned scores using its 7-point scale, known as the Clinical Global Impression-Severity scale (CGI-S). The CGI-S assesses the severity of the disorder, with lower scores indicating less severe symptoms. Within the Clinical Global Impression-Improvement scale (CGI-I), which evaluates changes in symptom severity over time, patients who received a rating of one or two on the CGI-S are regarded as experiencing clinically significant improvement, and as a result, they are categorized as responders. Conversely, individuals with a rating of three or higher on the CGI-S are classified as nonresponders, signifying a lack of substantial improvement in symptom severity. This comprehensive approach utilizing both CGI-S and CGI-I provides a nuanced evaluation of the patients’ overall condition and the effectiveness of interventions.

Iowa Gambling Task (IGT)

The Iowa Gambling Task (IGT) is a widely used test for assessing cognitive impulsivity. Participants are presented with four virtual card decks (A, B, C, and D) and must make choices to win or lose virtual money over 100 trials [26]. Some decks offer higher gains and losses, considered riskier choices, while others provide smaller gains and losses. To capture impulsivity, researchers often focus on the final 40 trials, as the initial trials involve a

learning phase. The IGT takes about 15 min to complete, and to prevent memory effects, two different versions are used, with the risky decks varied. Key measures include the percentage of risky choices and a net score indicating risk preference. Ekhtiari et al. developed the Persian version of the task at the Institute for the Study of Cognitive Sciences, which was proven to be reliable and valid when tested on Iranian participants [27, 28].

Tower of London Test (TOLT)

In this study, the computerized TOLT was administered using the Colorado Assessment Tests (CATs) for cognitive and neuropsychological assessment [29]. The TOLT assessed planning and problem-solving skills by requiring participants to move colored beads on a computer screen from a starting position to a goal position with as few one-by-one moves as possible [30, 31]. The test had 21 trials with varying numbers of colored beads on pegs, and performance measures were derived for 1,860 trauma-exposed individuals. These measures included average trial time (ATRTI) and excess moves (EM) for each puzzle type and were used across all puzzle types for this study. Data from the last TOLT evaluation were utilized, consistent with other study measures. In the Mowzoon study [32], the computerized Persian version of the TOLT was evaluated and found to have acceptable psychometric qualities, with a construct validity of approximately 0.41 and an internal consistency reliability (Cronbach’s alpha) of about 0.79 in Iranian samples.

The primary researcher, EZM, conducted the sessions and collected the data, with neither the patients nor the researcher being blinded.

Statistical analysis

Given the case-series design ($n=5$), analyses were descriptive only. We report means $\pm$ SD (or medians [IQR]), ranges, absolute change (Δ), and percent change where informative. No hypothesis testing or multiplicity corrections were performed. Per-case trajectories are provided in Supplementary files.

Results

Demographic data

A total of 35 participants from Ibn Sina hospital and 60 participants from Hafez hospital, all with an OCD diagnosis in their medical records, were assessed for our inclusion criteria and contacted about the study. Ultimately, five participants took part in the study and completed the intervention. In our study, the individuals diagnosed with OCD spanning from 18 to 46 years old. Among this group, four were female. Four patients had failed prior pharmacological regimens (≥ 1 SSRI and/or clomipramine with augmentation in several cases); one patient had failed structured CBT and was

Table 1 Overall characteristics of all participants

ID	Age	Sex	Age at diagnosis	Age of treatment starting	Duration of treatment (in years)	Prior treatment	Education	Number of children	Y-BOCS	Medications
1	23	Male	18	18	5	Pharmacological	University degree	NA	30	Aripiprazole, Bupropion, Sertraline
2	18	Female	14	15	4	Pharmacological	Diploma	NA	33	Clonazepam, Quetiapine, Pregabalin, Clo-mipramine, Fluoxetine, Mirtazapine
3	36	Female	7	26	29	Pharmacological	University degree	1	27	Sertraline, Risperidone
4	28	Female	13	14	15	Pharmacological	Diploma	2	23	Sertraline, Buspirone, Alprazolam, Quetiapine, Gabapentin, Melatonin
5	46	Female	38	45	8	CBT	Diploma	2	30	NA

Table 2 Dynamic changes in Y-BOCS over four weeks of the study

Variables	Start	Week 1	Week 2	Week 3	Final
Y-BOCS	28.4 (4.16)	24.6 (7.02)	22.4 (6.43)	20.8 (8.76)	18.4 (6.34)
Mean change from baseline (SD)	—	-3.8 (2.86)	-6.0 (2.27)	-7.6 (4.60)	-10.0 (2.18)
Percentage change	—	-13.4%	-21.1%	-26.8%	-35.2%

Abbreviations: Y-BOCS: Yale-Brown Obsessive-Compulsive Scale

Table 3 Changes in CGI-S, and CGI-I scores over the course of the study: a comparison of initial assessment and week 4

Variables	Start	Final	Percentage change
CGI_I		2.6	N/A
CGI_S	5.4 (0.89)	4.6 (1.14)	-14.8%

Abbreviations: CGI_I: Clinical Global Impression-Improvement scale; CGI_S: Clinical Global Impression-Severity scale

medication-naïve at baseline. Baseline Y-BOCS scores were ≥ 23 in all, consistent with at least severe symptomatology (Table 1).

Outcomes

Table 2 presents the progressive reduction in obsessive-compulsive symptom severity over four weeks. The mean baseline Y-BOCS score was 28.4 ± 4.16 , decreasing to 24.6 ± 7.02 after week 1, 22.4 ± 6.43 after week 2, 20.8 ± 8.76 after week 3, and 18.4 ± 6.34 at the final assessment. The mean change from baseline reached -10.0 ± 2.18 points, corresponding to an overall 35.2% reduction in symptom severity.

Table 3 summarizes the changes in clinical global impression scores across the study period. The mean CGI-S score decreased from 5.4 ± 0.89 at baseline to 4.6 ± 1.14 at week 4, reflecting a 14.8% reduction in illness severity. The mean CGI-I score at the end of treatment was 2.6, indicating “much improved” on average.

Supplementary Material, Table S1 summarizes the descriptive changes in affective and behavioral self-report measures over the four-week study period. The mean total DASS-21 score decreased from 38.8 ± 5.89 at baseline to 20.8 ± 13.99 at week 4, representing a 46.4% reduction. The Stress subscale declined from 16.8 ± 2.49 to 9.2 ± 6.34 (−45.2%), Depression from 13.6 ± 3.58 to 5.6 ± 5.81 (−58.8%), and Anxiety from 8.4 ± 4.83 to 6.0 ± 3.32 (−28.6%). The SBQ-R score also decreased modestly from 8.4 ± 5.81 to 7.0 ± 5.34 , corresponding to a 16.7% reduction in suicidal-behavior risk indicators.

Supplementary Material, Table S2 presents the descriptive changes in cognitive task performance between baseline and week 4. On the IGT, total gain increased from 2.0 ± 4.0 at baseline to 4.4 ± 13.45 at week 4 (+120.0%), while total scale decreased from 1465 ± 777.54 to 1290 ± 725.78 (−11.9%), and total time shortened from 339.56 ± 150.98 to 282.38 ± 89.84 (−16.8%). On the TOLT, time test decreased from 576.8 ± 419.14 to 409.4 ± 209.69 (−29.0%), time late from 268.4 ± 110.00 to 227.0 ± 98.83 (−15.4%), and time total from 845.2 ± 527.10 to 636.4 ± 269.04 (−24.7%). The result start score improved from 29.4 ± 1.82 to 31.2 ± 1.64 , reflecting a 6.1% enhancement in planning and executive performance.

Adverse effects

After completing all the sessions, the participants were requested to evaluate any negative consequences arising from the stimulation by providing self-reports. Thankfully, none of them experienced any noteworthy adverse outcomes. The most frequently mentioned effect was irritation, which was documented in four instances (cases 1, 3, 4, and 5), with three of them describing flashing sensations (cases 1, 2, and 4).

Discussion

The current exploration into the effectiveness of tACS for therapy-resistant OCD revealed noteworthy outcomes across several measures, particularly in clinical symptomatology. Our primary focus was on obsessive-compulsive symptoms, which, as assessed by the Y-BOCS, exhibited a trend toward reduction over the four weeks, with a mean change from baseline of -10.0 ± 2.18 points, corresponding to an overall 35.2% reduction in symptom severity. Given the high comorbidity of affective symptoms in therapy-resistant OCD, we also assessed emotional distress. This measure, the DASS-21, demonstrated a notable decline, with the mean total score decreasing by 46.4%. The most pronounced effect was observed in the depression subscale (a 58.8% reduction), while stress and anxiety levels also exhibited considerable reductions (45.2% and 28.6%, respectively). This underscores the broad positive impact of tACS on OCD associated psychopathology. The Clinical Global Impression Scale (CGI) further emphasized a favourable shift, with the mean CGI-S score decreasing by 14.8% and the mean final CGI-I score of 2.6 indicating that participants were “much improved” on average. Our secondary, exploratory measures yielded more varied results. Insights into suicidal tendencies (SBQ-R) showed a modest 16.7% reduction. Changes were also observed in cognitive task performance. On the Iowa Gambling Task, total gain increased (+ 120.0%) while total time decreased (– 16.8%). On the Tower of London test, performance was enhanced, with the result start score improving by 6.1% and total time decreasing by 24.7%. The safety of tACS was demonstrated in this study and was aligned to previous claims on its safety [13, 33].

Aligning with the promising effects of Transcranial magnetic stimulation in addressing OCD symptomatology [19, 34], the broader field of non-invasive brain stimulation has been systematically evaluated for OCD management [19]. Within this field, Transcranial electric stimulation (TES), commonly employed as direct current stimulation or tDCS was brought to the limelight of active investigation in treating psychiatric disorders such as treatment-resistant OCD. Multiple systematic review and meta-analysis suggest that obsessive-compulsive disorder can benefit from TES in the form of tDCS [20, 35]. However, investigations on the effects of tACS is fewer and non-conclusive.

Our findings on clinical symptoms align with earlier research, including the first reported case series by Klimke et al. [14] and a subsequent case study by Haller et al. [13], who both reported positive clinical effects. Our use of gamma-tACS over the bilateral DLPFC aligns with their approach; however, our protocol consisted of eight sessions, which is shorter duration than the 8–20 sessions they have administered. This difference in

treatment length may account for the differences in the outcomes compared to their studies, which reported significant clinical effects. Our study found a 35.2% reduction in Y-BOCS scores, which, while clinically noteworthy, must be interpreted in the context of our small sized, and open-label design.

Regarding participant characteristics, our study focused on a therapy-resistant OCD population, similar to the patients in the Klimke et al. case series. This differs from the non-clinical sample with obsessive-compulsive behaviors studied by Grover et al. [36], who found significant reductions after only five days. This suggests that a higher ‘dose’ of stimulation may be required to achieve effects in a more chronic, treatment-resistant population.

Finally, while our cognitive outcomes were varied and inadequate for performing significance tests, our finding of improved depressive symptoms (a 58.8% reduction on the DASS-21 Depression subscale) mirrors the report by Haller et al. [13], who also focused on clinical symptoms and showed some improvement in both OCD symptomatology and associated depressive features. This aligns with other studies showing the efficacy of tACS on the DLPFC for major depression [33, 37], suggesting our gamma-tACS protocol may have more robustly modulated the neural circuits shared between OCD and depression in our limited-duration trial.

tACS as a form of non-invasive Brain stimulation, is a low-cost, feasible and promising but a nascent tool in the field of psychiatry. The mechanism of action of tACS is the subject of current investigations. However, the proposed hypotheses behind its effectiveness are mostly to reinforce or reduce selective brain activities by entraining cortical neurons oscillations in selective frequency ranges and thus modulating connectivity to other brain regions by coupling or decoupling of these oscillations [38, 39]. Moreover, the after effects of tACS is regarded to be associated to spike-timing-dependent-plasticity (STDP) achieved by alternating the neuronal membrane potential and spike temporal patterns [12, 40].

Our study’s protocol was informed by the established link between OCD and dysfunctional dynamics within Cortico-Striato-Thalamo-Cortical (CSTC) circuits [41, 42]. This pathophysiology is thought to involve circuit hyperactivity, elevated low-frequency power, reduced alpha power, and distorted high-frequency oscillations in frontal areas linked to impaired reward processing [43, 44]. Therefore, our protocol targeted the dorsolateral prefrontal cortex (DLPFC), a key executive control node within this circuit [45, 46].

The choice of a gamma frequency was based on the specific hypothesis that enhancing high-frequency rhythms at the DLPFC could help modulate this pathological network activity and restore top-down control [13, 36, 47].

Therefore, our protocol is not just aimed at general brain stimulation, but at a targeted ‘tuning’ of a specific control node (the DLPFC) to a healthier operational frequency (gamma), with the downstream goal of re-balancing the broader pathological network activity that drives OCD symptoms. However, we must emphasize that, as this was a purely clinical case series without neurobiological measures (such as EEG or imaging), we cannot confirm that these intended mechanisms were engaged.

While our pilot investigation offers promising insights into the potential effectiveness of tACS in addressing therapy-resistant OCD, it is imperative to acknowledge and address certain methodological constraints. The small sample size employed in this study limits the generalizability of our findings and introduces a potential for selection bias. Participants who volunteer for a novel, time-consuming neuromodulation trial may possess a level of motivation, hope, or specific illness characteristics that are not representative of the broader treatment-resistant OCD population. Moreover, the absence of a control group poses challenges in attributing observed changes exclusively to the tACS intervention. This issue is critically compounded by the risk of measurement bias. For instance, active tACS can induce sensory phenomena, such as the perception of flickering lights (phosphenes) from retinal stimulation or a tingling sensation on the scalp. Without a convincing sham condition that mimics these sensations, participants and clinicians cannot be effectively blinded. This awareness of receiving an active treatment can profoundly influence subjective symptom reports, making it difficult to disentangle the true therapeutic effect from expectation.

Critically, as this was a clinical case series, we did not acquire EEG recordings, limiting the investigation of the quantitative electrophysiological benefits of the intervention and lack of evidence for the proper target engagement. The lack of neurobiological measures means the underlying mechanisms of the observed clinical changes cannot be determined from this study. Additionally, the relatively short follow-up duration in our study bounded the interpretability of the enduring effects of tACS. In regard to interpreting the results of this and other similar studies it is worth considering that placebo or non-specific effects on treatment-resistant individuals may be more substantial and need highly rigorous study design to parse.

In light of these limitations, future research studies should contemplate randomized controlled designs with expanded and diverse participant cohorts, integration of sham stimulation conditions, and protracted follow-up intervals. Furthermore, probing into individual variabilities in treatment responsiveness and refining stimulation protocols based on personalized neurobiological markers may contribute to the optimization of

tACS interventions, thereby augmenting their clinical efficacy in treatment-resistant OCD.

The positive reductions identified in obsessive-compulsive symptomatology (35.2% Y-BOCS reduction) and emotional distress (46.4% DASS-21 reduction) suggest promising prospects for this neuromodulatory approach. The notable impact on the depression substrate (58.8% reduction), as well as the broader positive influence on associated psychopathology, holds implications for refining treatment strategies tailored to address specific symptom domains within OCD. The clinical improvements demonstrated by the positive shift in the Clinical Global Impression Scale (CGI-I of 2.6, “much improved”) underscores the potential applicability of tACS. However, the modest changes in suicidal tendencies (16.7% reduction) and the non-conclusive results from our exploratory cognitive tasks prompt a cautious approach, urging further research to delineate the specific conditions and populations that may benefit most from tACS. As the study paves the way for future investigations, its findings invite researchers and clinicians to consider the nuanced implications of tACS in enhancing therapeutic outcomes for therapy-resistant OCD.

Conclusions

Our pilot study on the application of Transcranial Alternating Current Stimulation (tACS) in therapy-resistant OCD has provided encouraging preliminary findings and demonstrate its safety. Limitations, including the small sample size, lack of randomisation and control in the study design and the relatively short follow-up duration, underscore the need for cautious interpretation and further investigation. Moving forward, it is imperative to conduct larger-scale, controlled trials with diverse cohorts to validate and extend our initial observations. The incorporation of sham stimulation conditions and an extended follow-up period will contribute to a more nuanced understanding of the sustained therapeutic effects of tACS. Moreover, future research should explore individual differences in treatment response and assess neurophysiological outcomes to be able to refine stimulation protocols based on personalized neurobiological markers. Despite these challenges, our study lays a foundation for future endeavours in the realm of neuromodulation for therapy-resistant OCD, urging continued exploration and refinement of tACS interventions with the ultimate goal of advancing novel and effective treatment options for this challenging clinical population.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12888-026-07838-z>.

Supplementary Material 1

Supplementary Material 2

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Author contributions

RM and EZ lead the initial investigation, MH and ES evaluate and integrate findings, with RM and EZ drafting the initial version. ES handles the analysis, while oversight is maintained by BO, ZZ, and EH. All authors approve the final version for publication.

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Data availability

All information required is given in the text and supplementary materials, other supplementary information can be obtained upon email from the corresponding author.

Declarations**Ethics approval and consent to participate**

The study obtained consent from patients and guardians for publishing deidentified clinical information. Efforts will be made to conceal identities, though anonymity isn't guaranteed. Informed consent forms, completed in Persian, adhered to ethical standards in the Declaration of Helsinki. The study was approved by the Shiraz University of Medical Sciences (SUMS) under the code [IR.SUMS.REC.1402.094].

Consent for publication

Written informed consent was obtained from all patients, and no identifiable patient information is included in this manuscript.

Competing interests

The authors declare no competing interests.

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